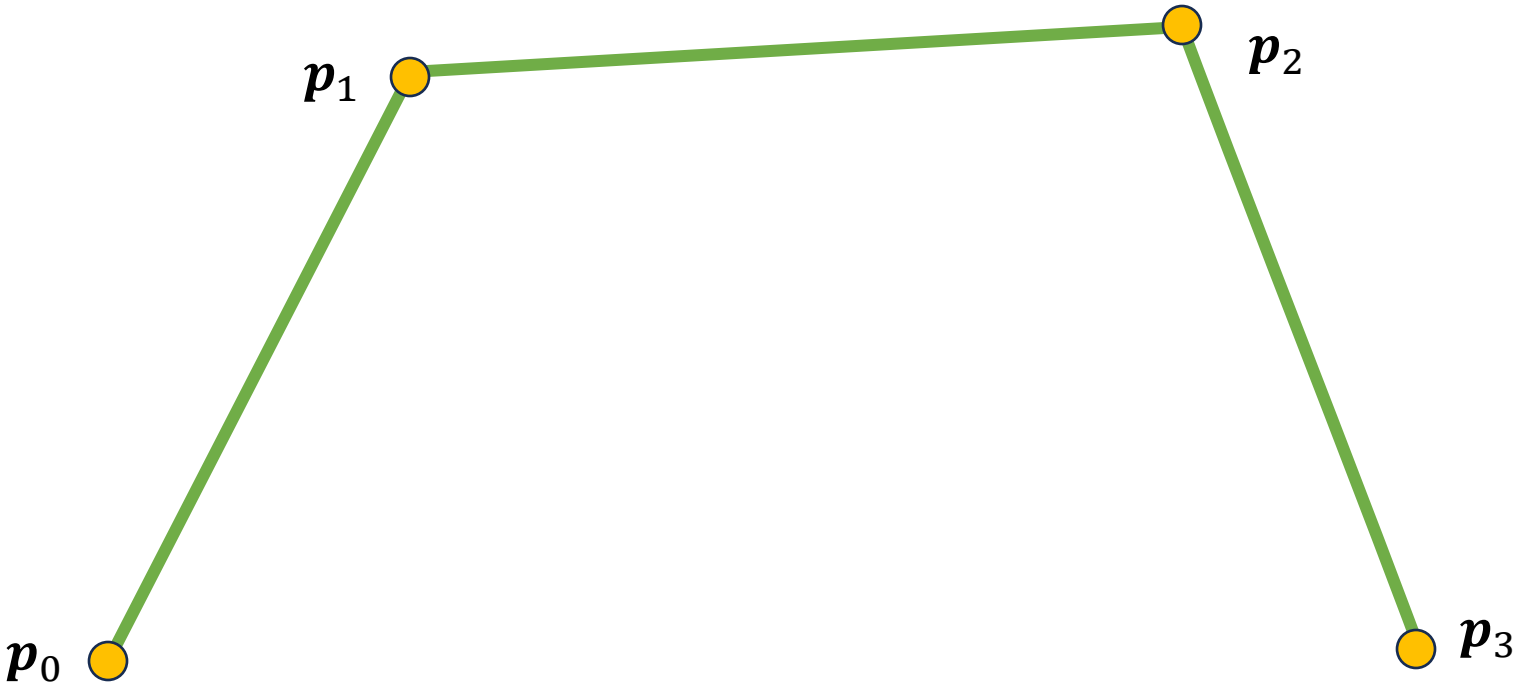
Subdivision Curves and Surfaces

SNU Computer Science & Engineering

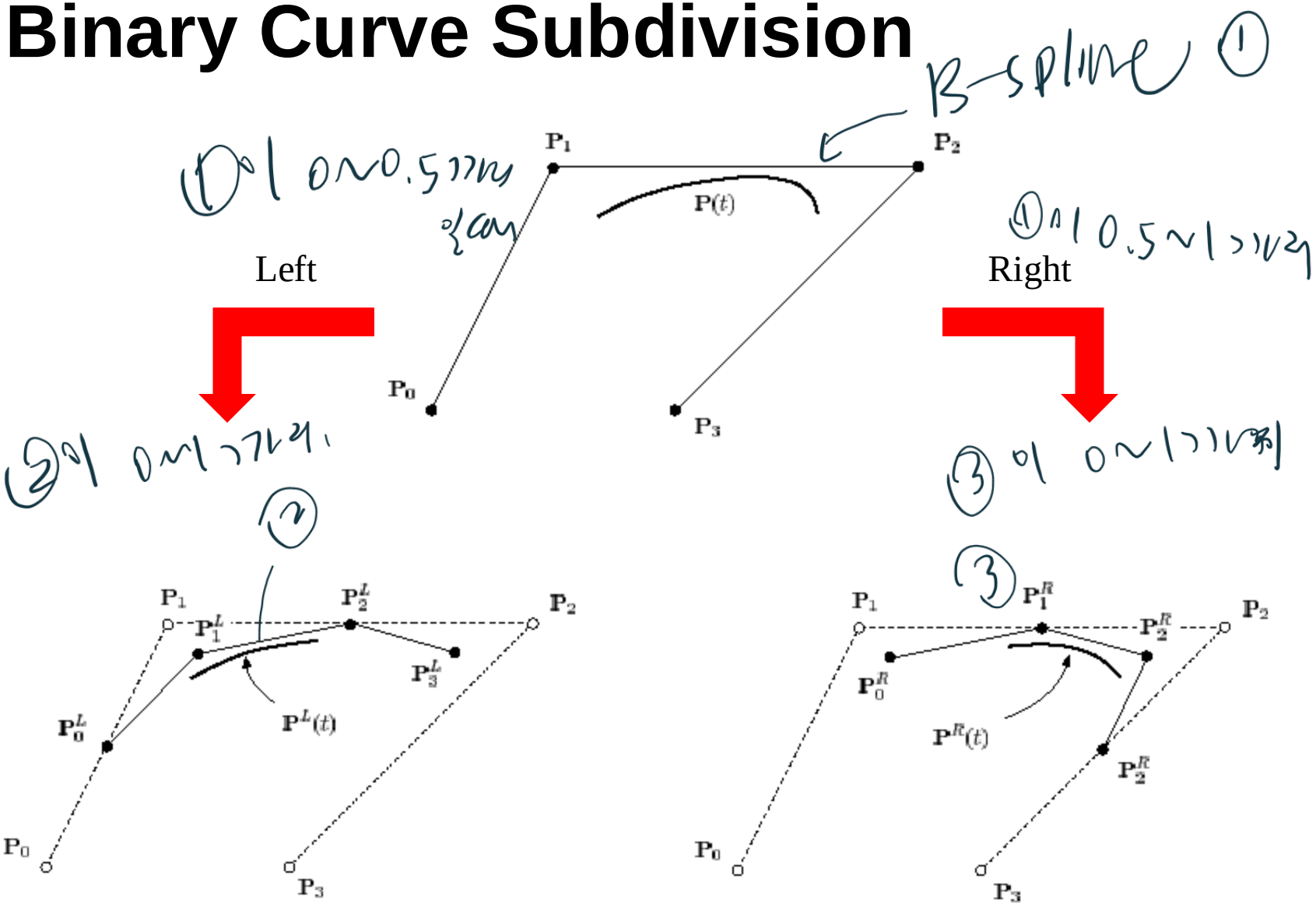
Some contents are stolen from the lecture slides of MIT Computer Graphics courses taught by Justin Solomon

Subdivision Curves

- Idea: Start with polygonal mesh, subdivide into larger number of polygons
- Control points eventually converge to underlying smooth curve (i.e., limit curve)



Binary Curve Subdivision



Binary Curve Subdivision

- The left segment

$$\boldsymbol{p}^L(t) = \boldsymbol{p}\left(\frac{t}{2}\right)$$

$$= [\boldsymbol{p}_0 \quad \boldsymbol{p}_1 \quad \boldsymbol{p}_2 \quad \boldsymbol{p}_3] \cdot \mathrm{B} \cdot \begin{bmatrix} 1 \\ t/2 \\ t^2/4 \\ t^3/8 \end{bmatrix}$$

$$= [\boldsymbol{p}_0 \quad \boldsymbol{p}_1 \quad \boldsymbol{p}_2 \quad \boldsymbol{p}_3] \cdot \mathrm{B} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{8} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix}$$

Binary Curve Subdivision

$$\boldsymbol{p}^L(t) = [\boldsymbol{p}_0 \quad \boldsymbol{p}_1 \quad \boldsymbol{p}_2 \quad \boldsymbol{p}_3] \cdot \mathrm{B} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{8} \end{bmatrix} \cdot \mathrm{B}^{-1} \cdot \mathrm{B} \cdot \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix}$$

$$= [\boldsymbol{p}_0^{\mathrm{L}} \quad \boldsymbol{p}_1^{\mathrm{L}} \quad \boldsymbol{p}_2^{\mathrm{L}} \quad \boldsymbol{p}_3^{\mathrm{L}}] \cdot \mathrm{B} \cdot \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix}$$

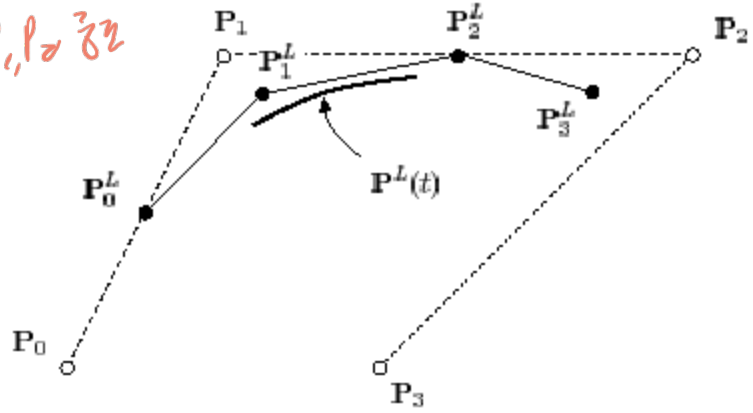
새로운 control point

= new B-spline

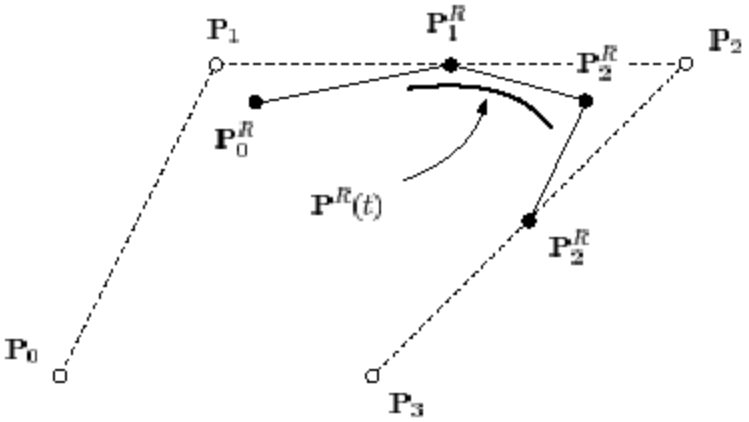
$\boldsymbol{p}^R(t)$ can be obtained by setting $\boldsymbol{p}^R(t) = \boldsymbol{p}\left(\frac{t+1}{2}\right)$

Binary Subdivision for Cubic B-Spline

$$\begin{bmatrix} \boldsymbol{p}_0^L \\ \boldsymbol{p}_1^L \\ \boldsymbol{p}_2^L \\ \boldsymbol{p}_3^L \end{bmatrix}^{\mathrm{T}} = \left(\begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/8 & 6/8 & 1/8 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 1/8 & 6/8 & 1/8 \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_0 \\ \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \\ \boldsymbol{p}_3 \end{bmatrix} \right)^{\mathrm{T}}$$



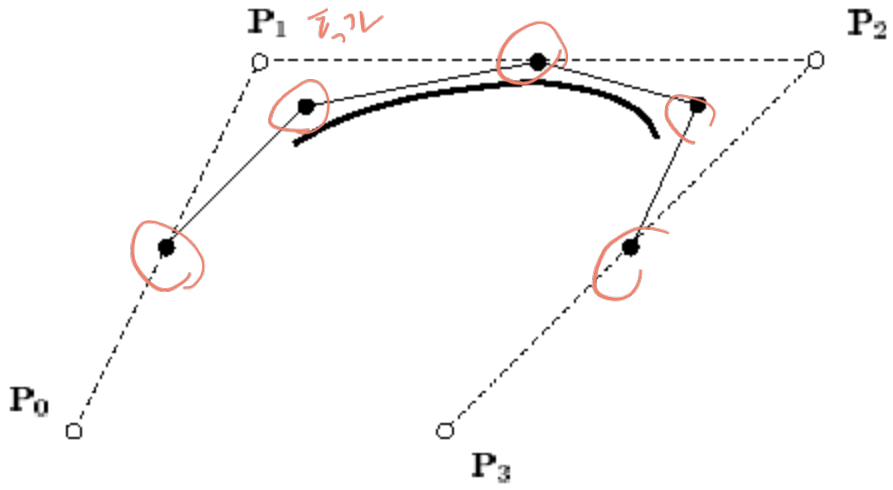
$$\begin{bmatrix} \boldsymbol{p}_0^R \\ \boldsymbol{p}_1^R \\ \boldsymbol{p}_2^R \\ \boldsymbol{p}_3^R \end{bmatrix}^{\mathrm{T}} = \left(\begin{bmatrix} 1/8 & 6/8 & 1/8 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 1/8 & 6/8 & 1/8 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_0 \\ \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \\ \boldsymbol{p}_3 \end{bmatrix} \right)^{\mathrm{T}}$$



Subdivision Rule for Cubic B-Splines

- The new control polygon consist of
  - **Edge points**: the midpoin (1:1) of each line segment
  - **Vertex points**: the weighted average (1:6:1) of the corresponding vertex and its two neighbors

$$\begin{bmatrix} \boldsymbol{p}_0^1 \\ \boldsymbol{p}_1^1 \\ \boldsymbol{p}_2^1 \\ \boldsymbol{p}_3^1 \\ \boldsymbol{p}_4^1 \end{bmatrix}^T = \left(\begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/8 & 6/8 & 1/8 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 1/8 & 6/8 & 1/8 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_0 \\ \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \\ \boldsymbol{p}_3 \end{bmatrix} \right)^T$$



Subdivision Rule for Cubic B-Splines

- ***Recursive subdivision*** brings the control polygon to converge to a cubic B-spline curve
- Control polygon + subdivision rule
  - Yet another way of defining a smooth curve



Can we find other subdivision rules?

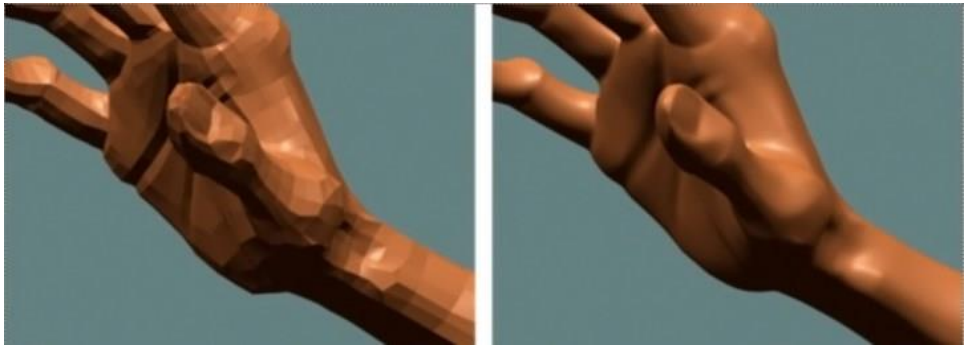
Landmark in Subdivision/Cloth

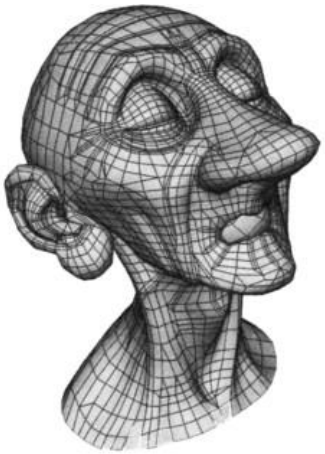
- ***Geri's Game***, Pixar 1997



Landmark in Subdivision/Cloth

- ***Geri's Game***, Pixar 1997





(a)



(b)

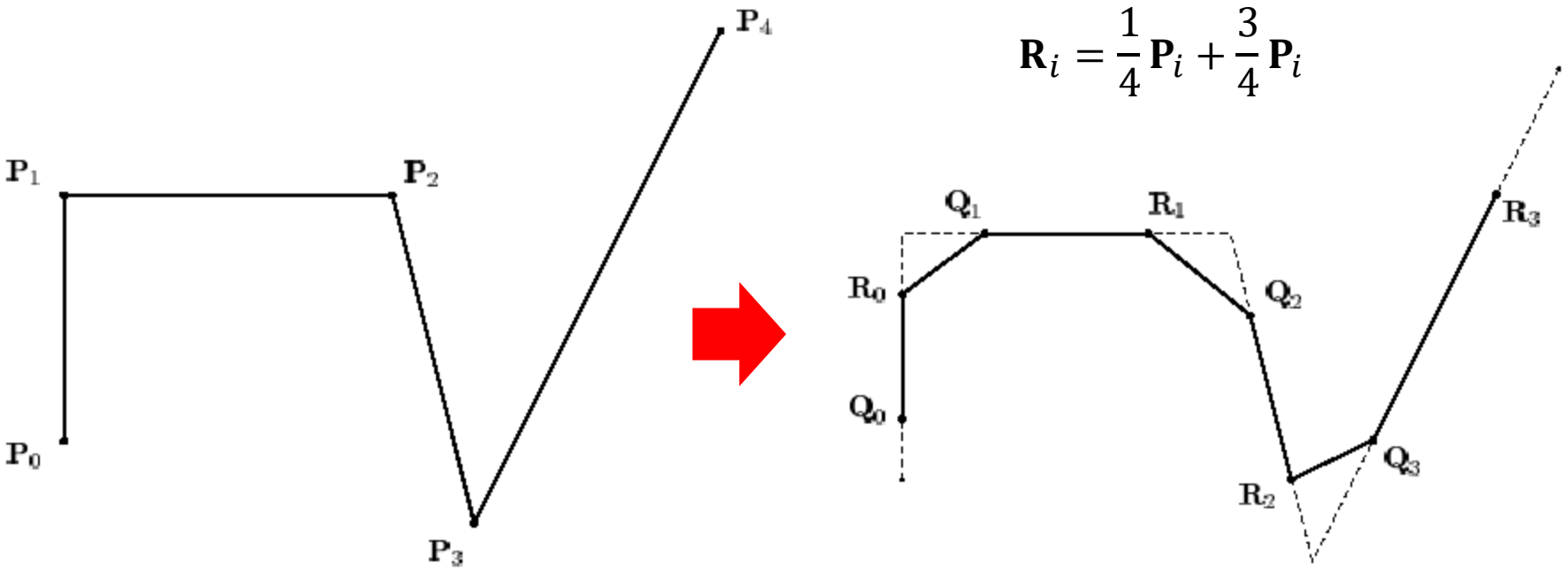
Walter Meyer, in Geometry and Its Applications (Second Edition), 2006

Chaikin's Algorithm

- *Coner cutting* (non-stationary subdivision)
- Converges to a quadradic B-Spline curve

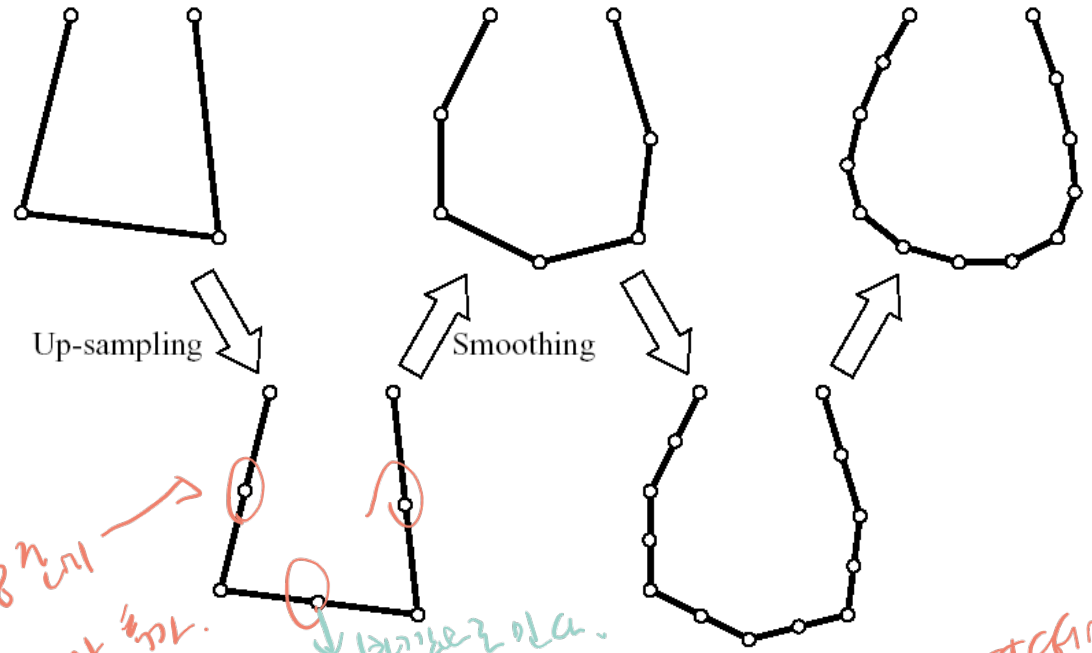
$$\mathbf{Q}_i = \frac{3}{4}\mathbf{P}_i + \frac{1}{4}\mathbf{P}_i$$

$$\mathbf{R}_i = \frac{1}{4}\mathbf{P}_i + \frac{3}{4}\mathbf{P}_i$$



Interpolating (Stationary) Subdivision

- The original control points are interpolated (remained)
  - ***Vertex points***: The original control points
  - ***Edge points***: The weighed average of the original points

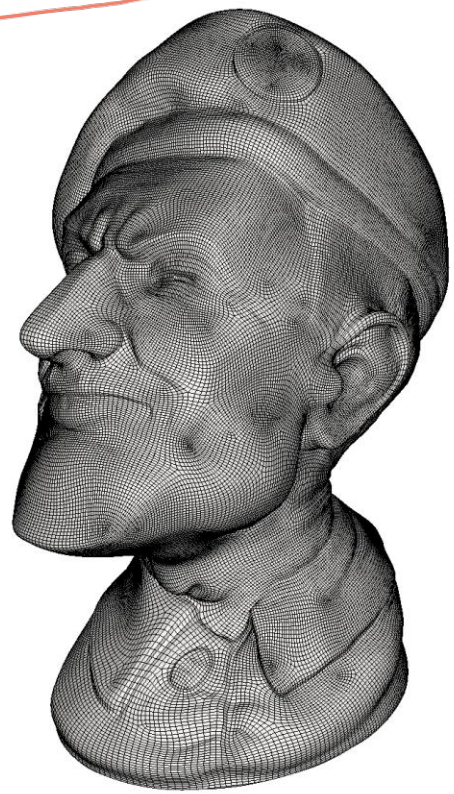


$$\mathbf{p}_{2i+1}^{m+1} := \frac{1}{16}(-\mathbf{p}_{i-1}^{m} + 9\mathbf{p}_{i}^{m} + 9\mathbf{p}_{i+1}^{m} - \mathbf{p}_{i+2}^{m}).$$

Subdivision for Surfaces



Subdivision for triangular mesh:
Loop Subdivision

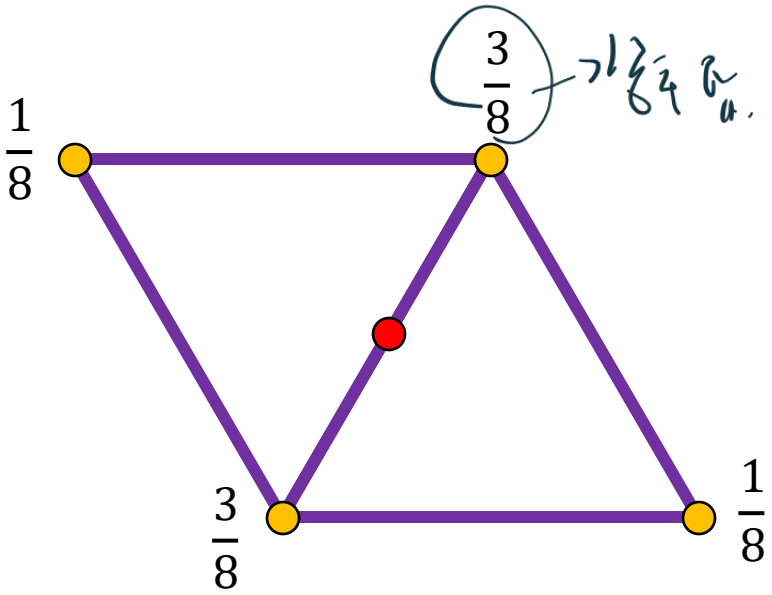


Subdivision for arbitary mesh:
Catmul-Clark Subdivision

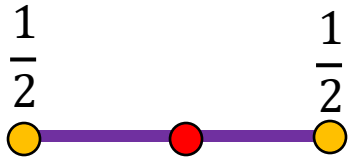
C^2 almost everywhere (C^1 at extraordinary vertices)

Loop Subdivision [Loop 1987]

- STEP1
  - For each edge, add an **edge point**



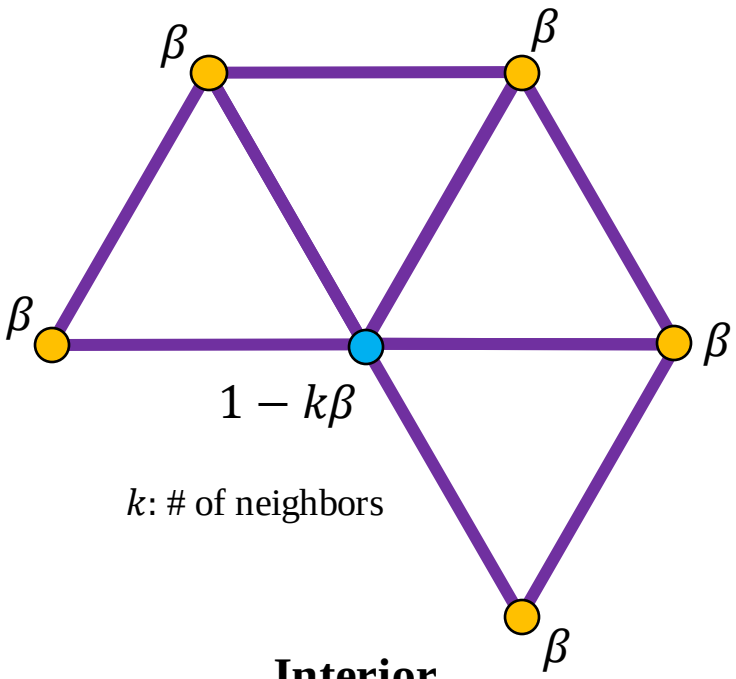
Interior



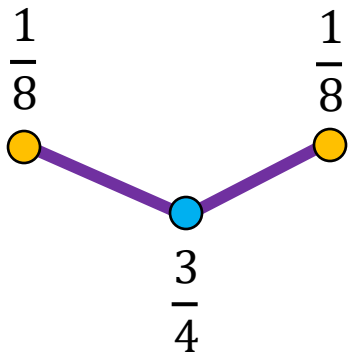
Boundary

Loop Subdivision [Loop 1987]

- STEP2
  - For each vertex, move it to **new position**



Interior



Boundary

Loop Subdivision [Loop 1987]

- How to choose β?

$$\beta = \frac{1}{n}\left(\frac{5}{8} - \left(\frac{3}{8} + \frac{1}{4}\cos\frac{2\pi}{n}\right)^2\right)$$

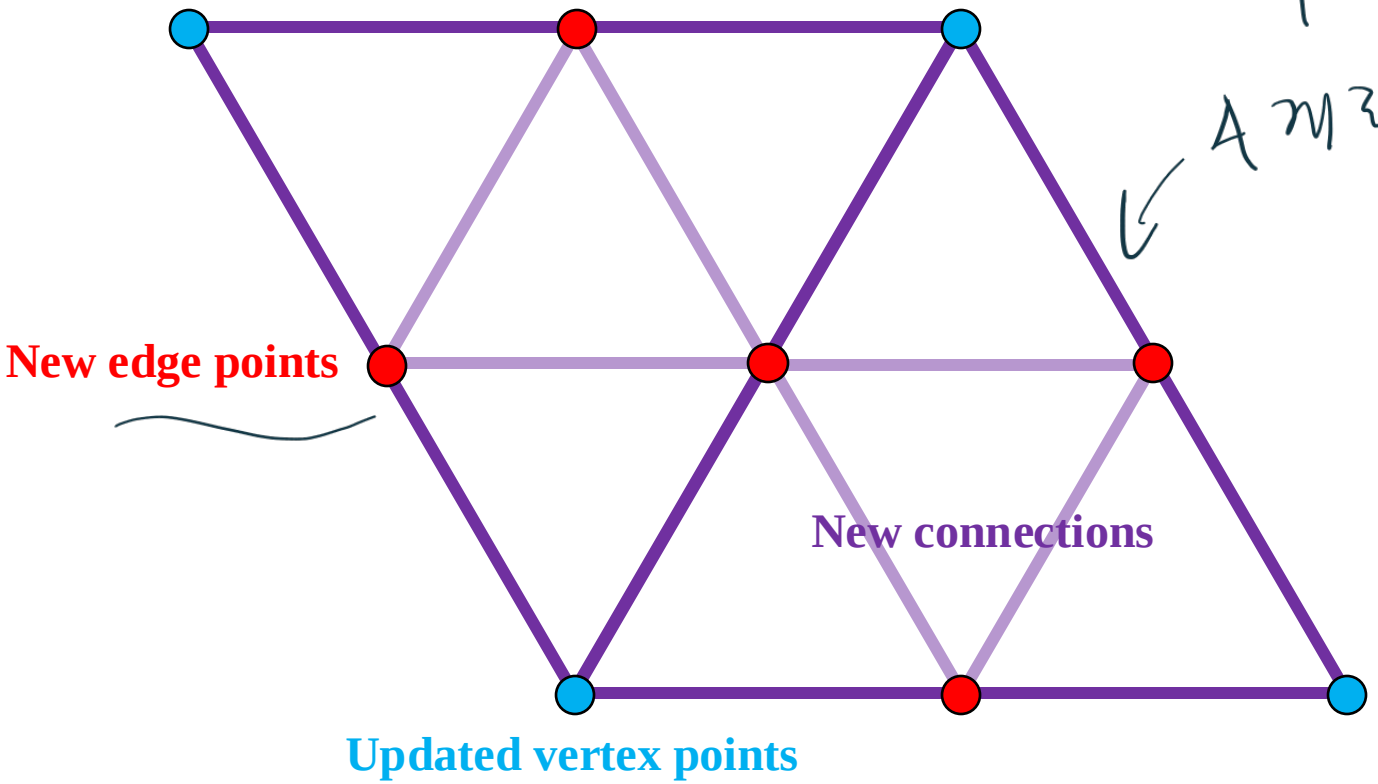
by [Loop 1987]

$$\beta = \begin{cases} \frac{3}{16} & \text{if } n = 3 \\ \frac{3}{8n} & \text{if } n > 3 \end{cases}$$

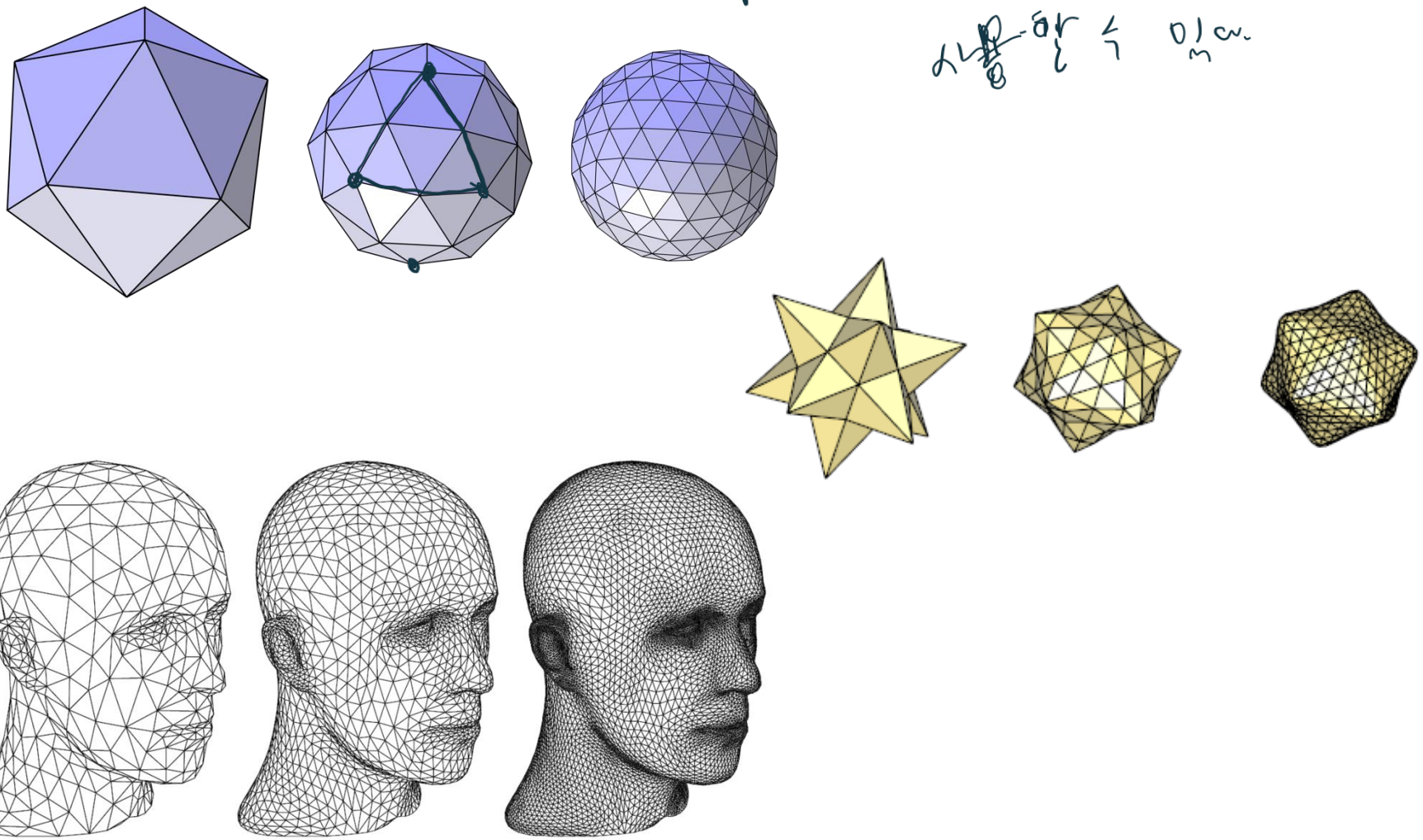
by [Warren 1995]

Loop Subdivision [Loop 1987]

- STEP3
  - Connect edge/vertex points accordingly

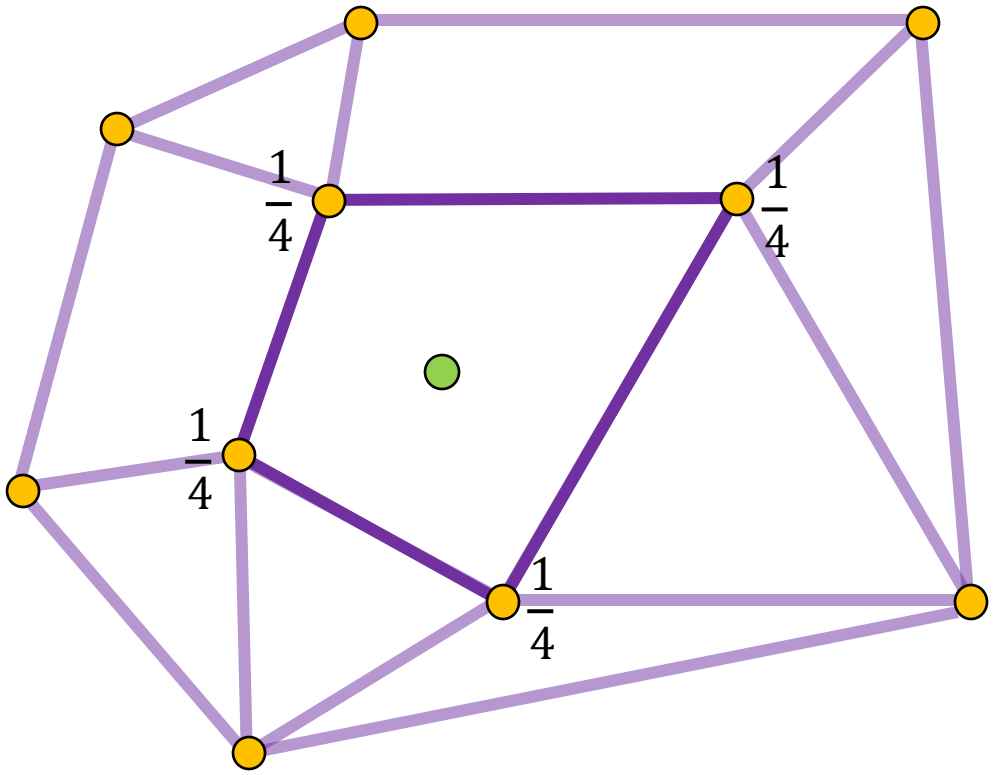


Loop Subdivision: Examples



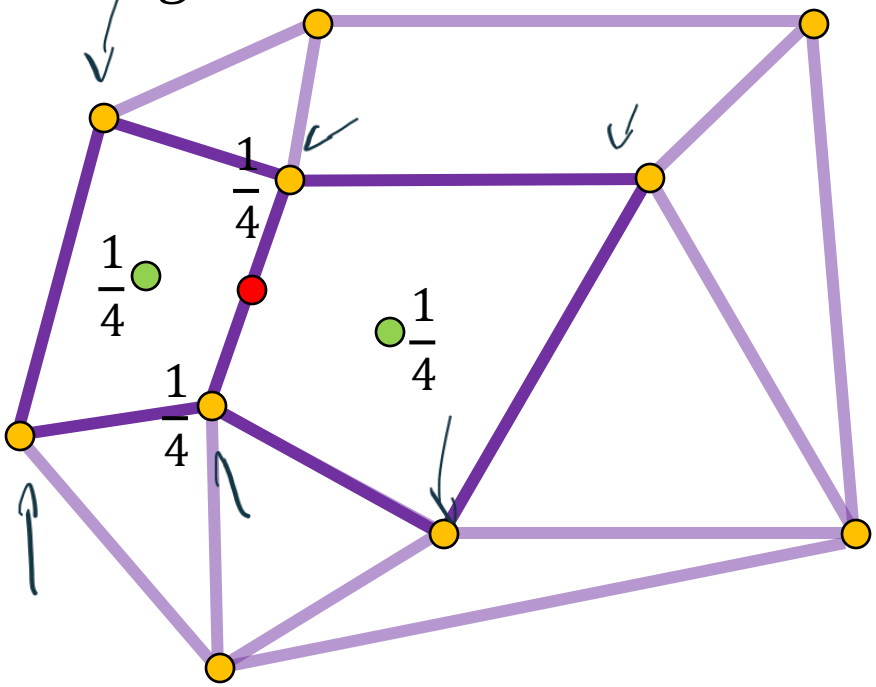
Catmull-Clark Subdivision

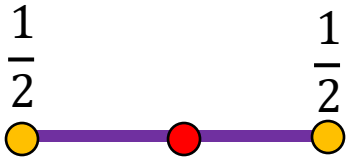
- STEP1
  - For each face, add a **new face point** which is the average of original vertex points



Catmull-Clark Subdivision

- STEP2
  - For each edge, add an **edge point** which is the average of the neighbouring face points and the two endpoints of the edge



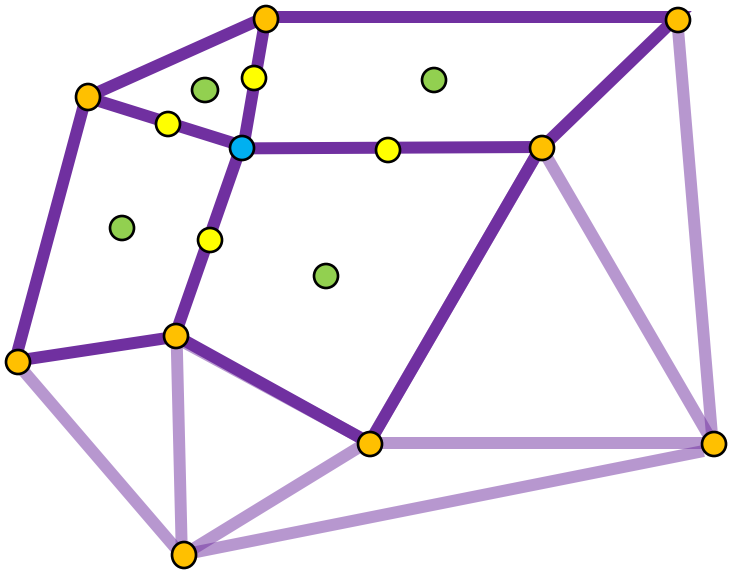


Interior **Boundary**

Catmull-Clark Subdivision

- STEP3
  - For each vertex, compute a **new vertex position** P' by taking the average of F, R, and P

Interior



$$P' = \frac{F + 2R + (n-3)P}{n}$$

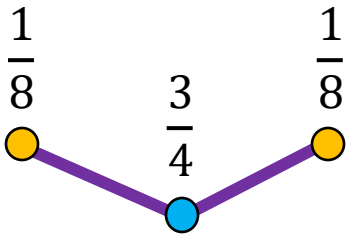
P: the original vertex point with valence n

F: the average of face points (green) for faces touching P

R: the average of edge midpoints (yellow) for edges touching P

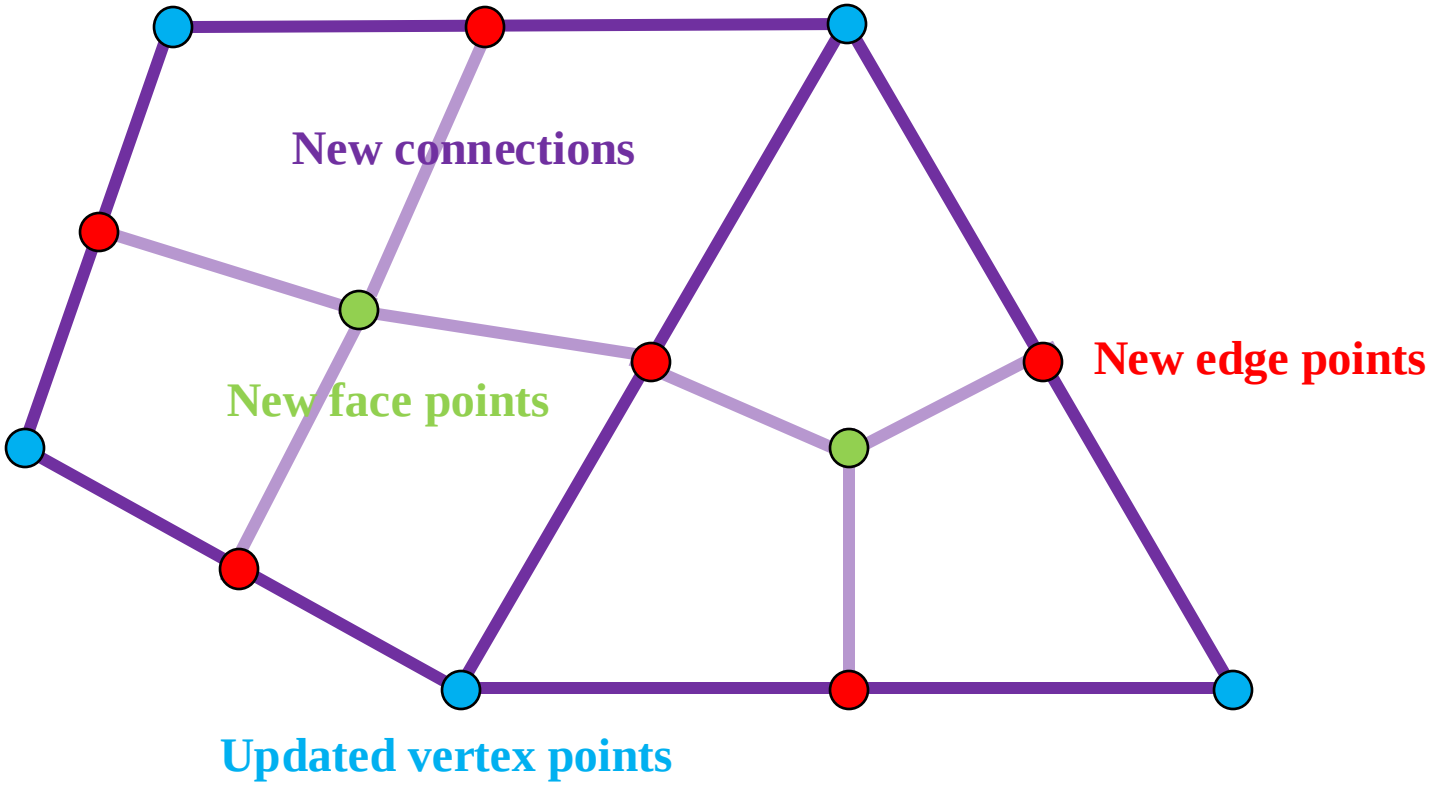
Note: Do not confuse with new edge points generated in step2

Boundary

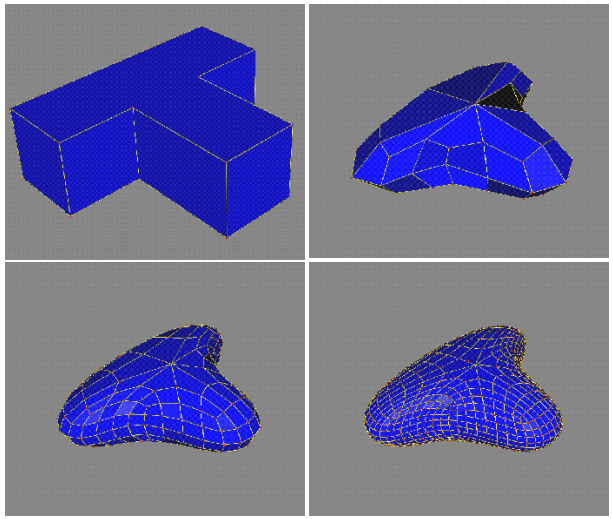


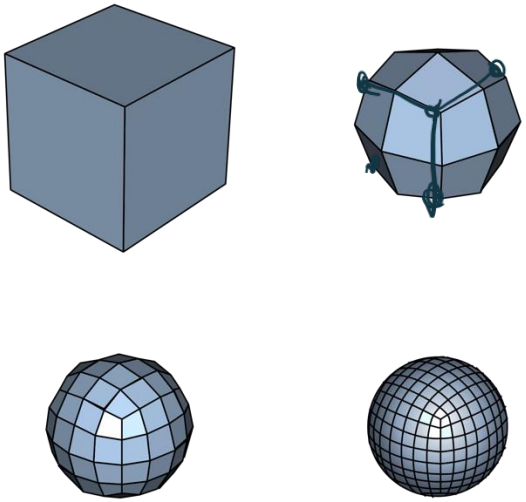
Catmull-Clark Subdivision

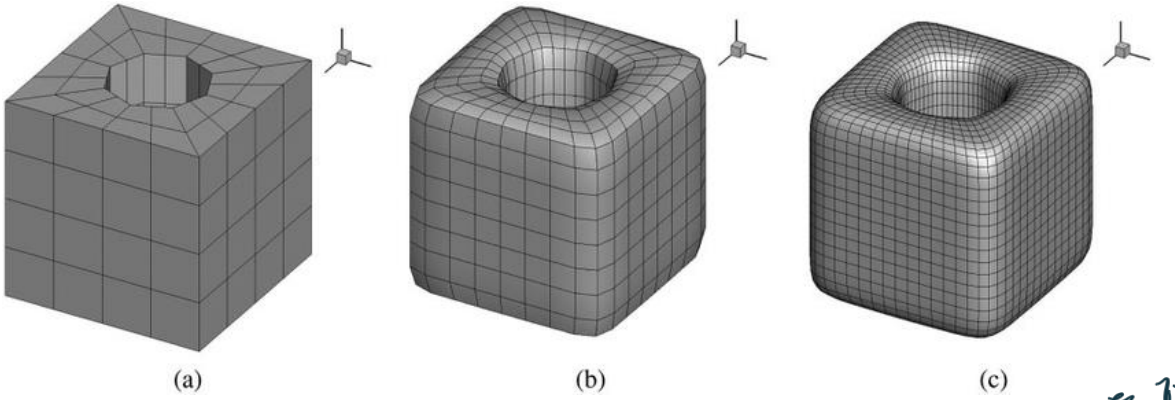
- STEP4
  - Connect edge/face/vertex points accordingly



Catmull-Clark Subdivision: Examples

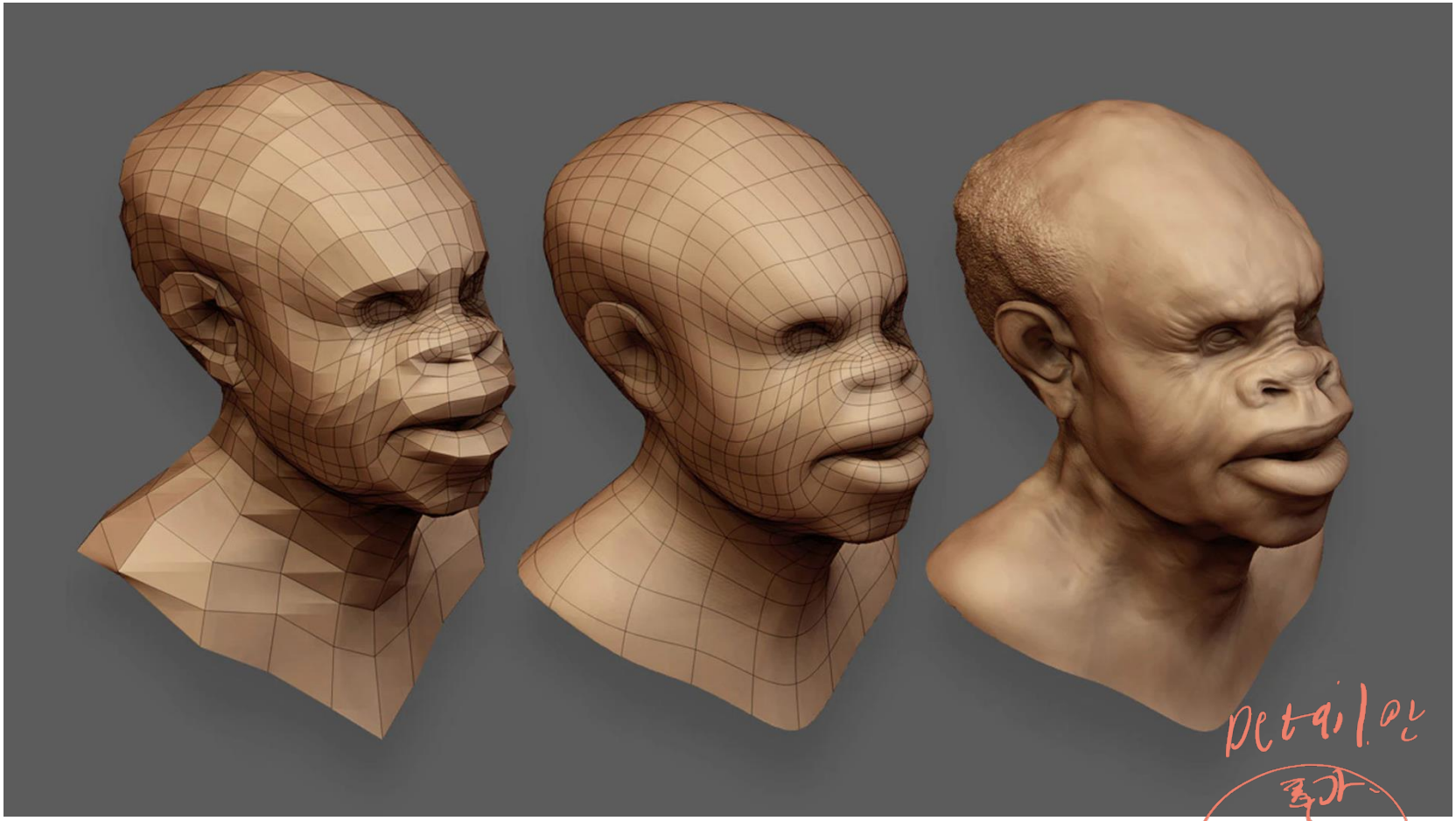






Level of detail (LOD) : close up → subdivision 많이 → 부드러움.
멀리 보면 → 〃 적게

Displacement Mapping (Later Topic)



Base Model Subdivision Subdivision + Displacement

Any Question?